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PI Name:	Gilroy, Simon Ph.D.		
Project Title:	The Interaction Of Touch And Gravitropic Signaling In Plant Roots		
Division Name:	Space Biology		
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Comments:	NOTE: PI formerly at Pennsylvania State University; moved to University of Wisconsin-Madison in 2007 (Info received 7/2009)		
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No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
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	Touch and gravity are two of the many stimuli that plants must integrate to generate an appropriate growth response. Due to the mechanical nature of both of these signals, shared signal transduction elements could well form the basis of the cross-talk between these two sensory systems. Two of the most widely proposed candidate signal transduction elements in both plant touch and gravity sensing are changes in Ca2+ and pH in the cytoplasm of the sensory cells. However, the precise spatial and temporal characteristics of changes in these signaling elements in response to mechanical and/or gravity stimulation, and especially how they might interact in a signaling cross-talk system, are poorly defined. The overall goal of this experimental program has therefore been to seek to define the spatial and temporal character of the ionic signaling associated with mechano-perception. Using this characterization we then mimicked or blocked the appropriate dynamics of these ionic changes and so helped define how they regulate the		

Task Description:	signaling systems responsible for the gravitropic response and generation of plant form. Touch stimulation has dramatic effects on plant morphogenesis and its interaction with gravimorphogenesis has a critical role in determining overall plant development and productivity. Thus, understanding the molecular mechanisms behind touch and gravity sensing systems and how these interact to generate plant form will not only contribute to our fundamental understanding of how plants grow, but may also suggest avenues for novel approaches to engineering crop stature or growth habit. In addition, it may be possible to manipulate the gravity response system of the plant with appropriate mechanical stimulation to circumvent problems of plant growth in the weightless environment of space flight.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research has employed state-of-the-art cell biological tools to elucidate the molecular and cellular mechanisms that translate the force of gravity into oriented growth of the plant. In addition, the research aimed to define how other signals perceived by the cell might impact on this regulatory system and the molecular mechanisms whereby plants exhibit touch sensitivity. An improved understanding of the regulation of plant gravity sensing mechanisms and how these are altered by the plant's environment should help design approaches to enhance plant growth in microgravity, providing increased plant productivity during sustained space flight and increased control over future spaceflight experiments.
Task Progress:	The primary goals of this research have been to determine the signaling components present in plant roots that allow for both gravity and touch sensitivity. In addition, we have sought to define how these elements may interact to allow the plant to integrate multiple stimuli to generate the most appropriate growth response to an ever-changing environment rich in multiple, often conflicting stimuli. We have found that gravity stimulation of plants leads to a rapid and transient increase in cellular pH in the sensory (columella) cells of the root tip. Although blocking this change does not completely block the gravity-related growth response of the root, it does severely curtail the degree of oriented growth. Thus, the pH change is part of a mechanism to enhance the oriented growth response, increasing the sensitivity of the plant to gravity stimulation. These pH changes are disrupted in mutants known to be compromised in aspects of gravity sensing, such as the starch deficient mutant pgm1 or the gravity insensitive mutant arg1, suggesting a functional role gravity response. However, mutants disrupted in the gene adenosine kinase are known to show reduced gravitropic response, yet the pH changes in these plants were unaltered relative to wild-type. Thus, there appear to be pH-dependent and –independent pathways for gravity signaling, both of which are required for normal gravitropic response. We have found that the normal gravity-induced pH changes and the subsequent gravitropic growth response are dependent on the actin cytoskeleton. Actin appears to act to reset the system to its non-stimulated state after an initial sensory event and subsequent signal transduction has been generated. Thus, pharmacological disruption of actin function with very low concentrations of the drug Latrunculin B led to sustained pH changes in the gravity sensing cells and a significantly enhanced growth response to gravity stimulation. These alterations in response led to an overshoot in the final direction adopted by the organ to the gravity vector. The levels of Latrunculin B used for this analysis did not alter growth rates, suggesting the actin-dependent machinery being targeted is specific to the gravity sensing signal transduction system. Touch stimulation of the root cap was found to inhibit the cytosolic pH changes associated with gravity signaling in the sensory cells of this organ. Further, touch stimulation elicited a Ca2+ increase in the peripheral cells of the root cap that propagated to the gravity sensing cells of the root cap columella. This Ca2+ wave is independent of the gravity sensing system, for example it is unaltered in pgm1 or arg1. Analysis of a library of mutants disrupted in all putative Ca2+ transporters in the plant Arabidopsis has revealed that this Ca2+ signal is sustained by the action of the Ca2+ transporter ACA12, providing a potential molecular target to manipulate mechanosenitivity in plants. We have also found that touch stimulation of the root surface caused not only a rapid and transient increase in cytosolic Ca2+, but also a rapid (within 1 s) localized and transient cell wall alkalinization and cytoplasmic acidification. Touch likewise elicited production of reactive oxygen species (ROS) in the cell wall localized to the area of mechanostimulation. This elevated ROS subsequently triggered transient alterations in the physical characteristics of the plasma membrane manifested as a decrease in affinity for the lipophilic fluorescent dyes such as FM 4-64. These responses could be elicited in the absence of touch by artificially increasing Ca2+ with the ionophore BrA23187 and were blocked by the Ca2+ channel inhibitor La3+. In the rhd2 Arabidopsis mutant that lacks a functional NADPH oxidase ATRBOH C, touch stimulation still triggered extracellular and cytoplasmic pH changes but not the local increase in ROS production, or ROS-dependent FM 4-64 disappearance seen in wild-type plants. Thus, touch responses likely elicit Ca2+-dependent activation of ATRBOH C yielding ROS production to the cell wall. This ROS production is coordinated with intra- and extracellular pH changes through the same touch-induced cytosolic Ca2+ transient. These localized changes in pH, ROS and membrane structure likely represent part of a rapid defense response to mechanical stress. These changes in wall environment should lead to the wall becoming rigidified due to covalent cross-linking by the ROS, activation of wall strengthening enzyme activities such as pectin methyl esterases and reduction in cell wall loosening activities such as those of the expansin gene family through the elevated pH. Such wall modifications likely play two roles in the plant: (1) as a defense response to local mechanical stimulation as experienced, for example, during pathogen or herbivore attack, and (2) as an intrinsic response to the expansive forces experienced by plant cells during normal growth. Thus, manipulation of this Ca2+-dependent extracellular regulatory system may allow modification of plant stress responses as well as allow a novel opportunity to regulate plant growth with cellular resolution. The growth responses elicited by both these touch- and gravity- responsive systems appear intimately connected to the plant hormone auxin. Thus, we have found that prolonging the gravity signal generated in the sensory cells of the root cap by disrupting actin, as described above, leads to a significant increase in auxin export from these sensory cells to those in the region of the root undergoing growth regulation. Therefore, in collaboration with Roberto Gaxiola (University of Connecticut), we have explored some of the driving forces for such auxin transport that operate in addition to the well-recognized roles of the PIN, PGP and AUX/LAX families of auxin transporters. We have determined that the vacuolar proton pumping pyrophosphatase (AVP1) also plays a role in root growth responses. This protein appears to sustain the hormonal fluxes required for integration of growth, likely through regulation of proton fluxes in the vesicles that correctly target the auxin transport systems and primary proton pumps of the plant to the plasma membrane. The novel role for this protein was revealed from analysis of growth and pH dynamics in the

	vacuole, cytoplasm and wall of plants disrupted in AVP1 production or over-expressing avp1. AVP1 therefore provides another molecular target for manipulation of plant growth responses to both the touch and gravity signal.
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